

Minimal-Shot Autonomous Driving

Geometry-Grounded Runtime, Transfer Diagnostics, WOD-E2E Grounding

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GitHub branch: `CoRL-2027`

This stack is built to handle unfamiliar long-tail traffic structure through explicit geometry, bounded actions, and auditable transfer tests.

Motivation

Autonomy that depends on collecting every rare case will always lag reality.

Minimal-shot autonomy needs a different capability:

Standard AV scaling	Minimal-shot target
train on many mapped cases	reason from first principles
optimize average-case driving	survive long-tail scenarios
hide failures inside aggregate metrics	expose failures by scenario and axis
expensive data dependence	reusable simulator and audit harness

The repo connects runtime, simulator, transfer harness, and real-data grounding in one reproducible pipeline.

System Contributions

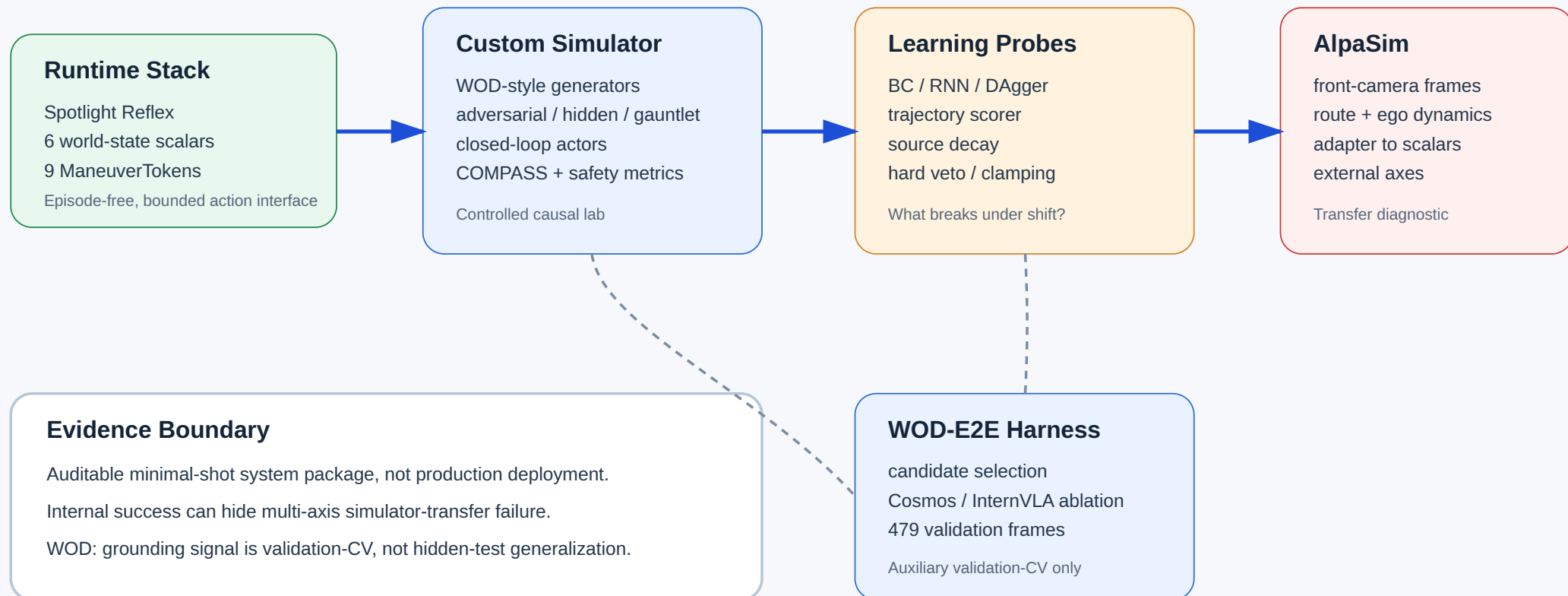
| Layer | What we built | Why it matters |
| --- | --- |
| runtime | Spotlight Reflex token policy | auditable geometry-first decision path |
| simulator | custom 2D long-tail environment | controlled causal stress testing |
| learning probes | BC, RNN, DAgger, scorer variants | isolates what imitation actually learns |
| transfer harness | AlpaSim proxy-state adapter | exposes external failure modes |
| grounding track | WOD-E2E selector harness | tests visual/world-model priors on real data |

The stack is useful because these layers connect into one evidence chain rather

Whole System Map

Minimal-Shot AV Stack

Runtime, simulator, transfer harness, and grounding probes share one bounded action interface.

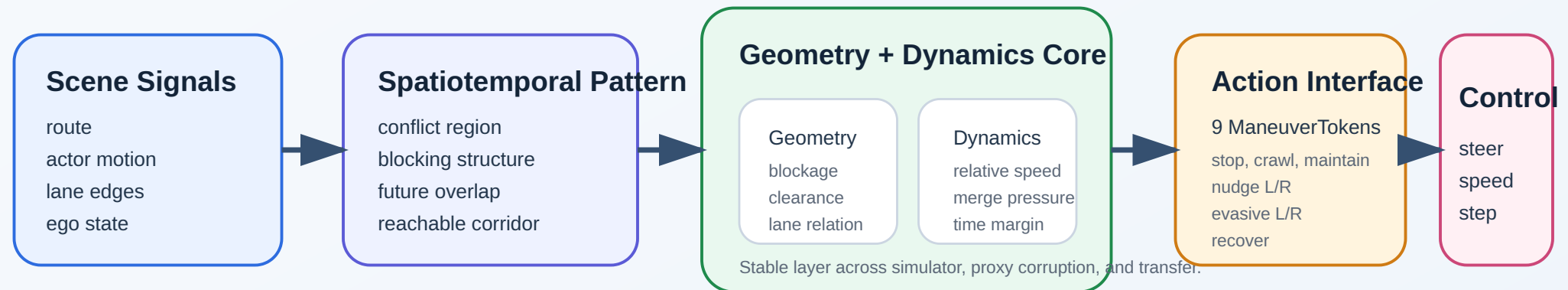


Runtime Architecture

FAST PATH FOR MINIMAL-SHOT DRIVING -> SPATIOTEMPORAL PATTERN -> GEOMETRY -> DYNAMICS -> ACTION

Grounded action emerges from spatiotemporal structure.

Compress motion first, resolve geometry and dynamics second, keep the action space bounded at the last step.



Bounded action keeps the last step explicit and auditable.

Token Interface

The same action interface is used by the rule policy, learned policies, and AlpaSim transfer experiments.

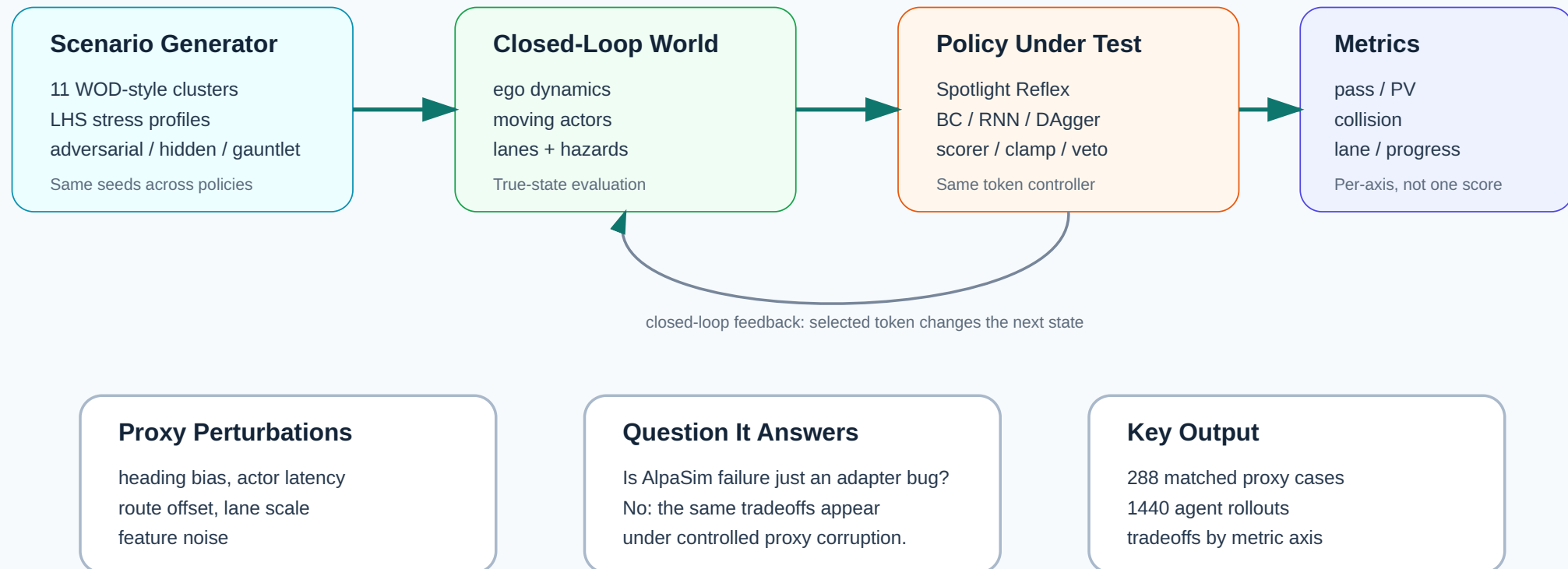
Token family	Examples	Role
longitudinal safety	stop, crawl, slow_yield	reduce collision pressure
nominal progress	maintain	continue along route
mild lateral evasion	nudge_left, nudge_right	create local clearance
emergency evasion	evasive_left, evasive_right	handle close hazards
recovery	lane_recover	return to lane center

This shared token interface makes failures diagnosable: we can separate bad candidate geometry from bad token selection.

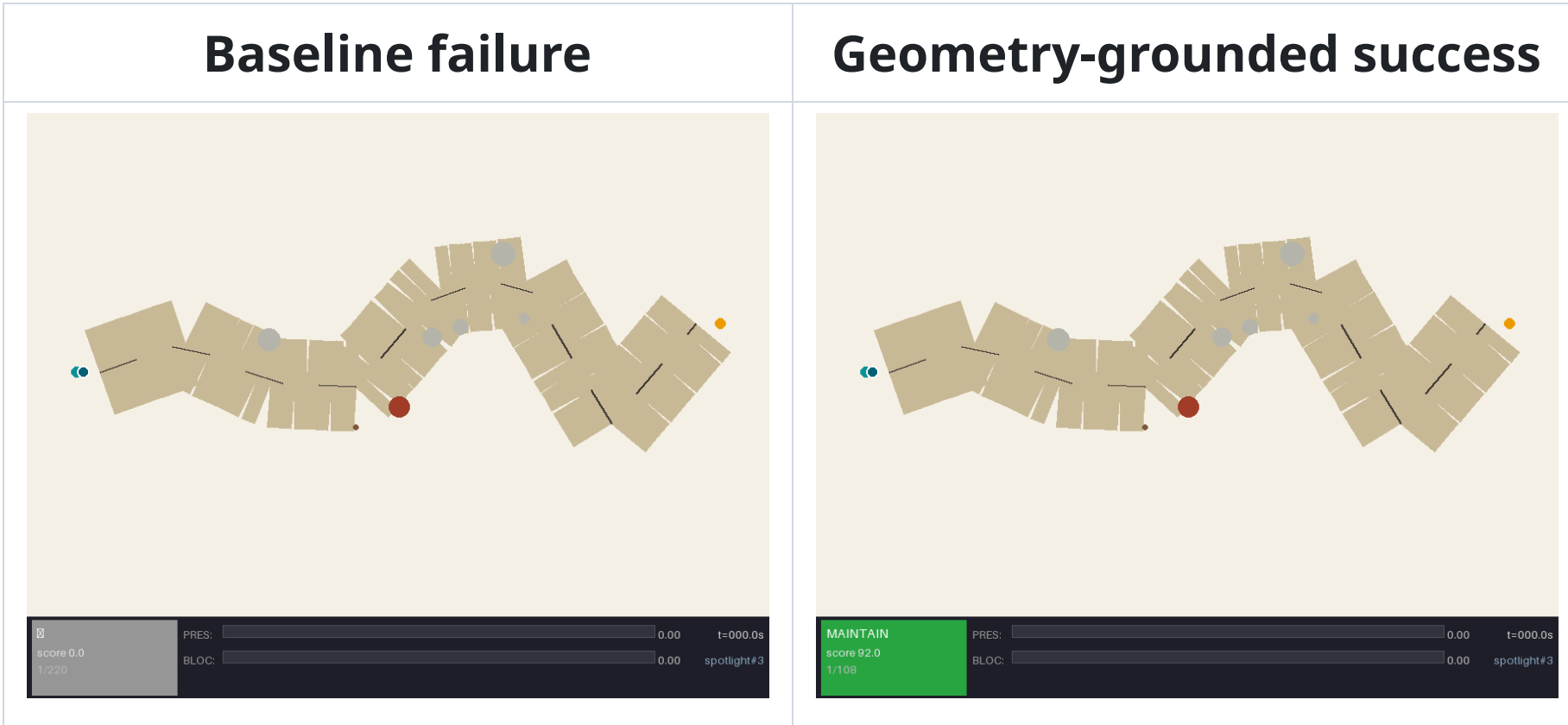
Simulator Environment

Custom Simulator: Why It Exists

It is a controlled stress lab where state, proxy corruption, policy, and metrics are independently manipulable.



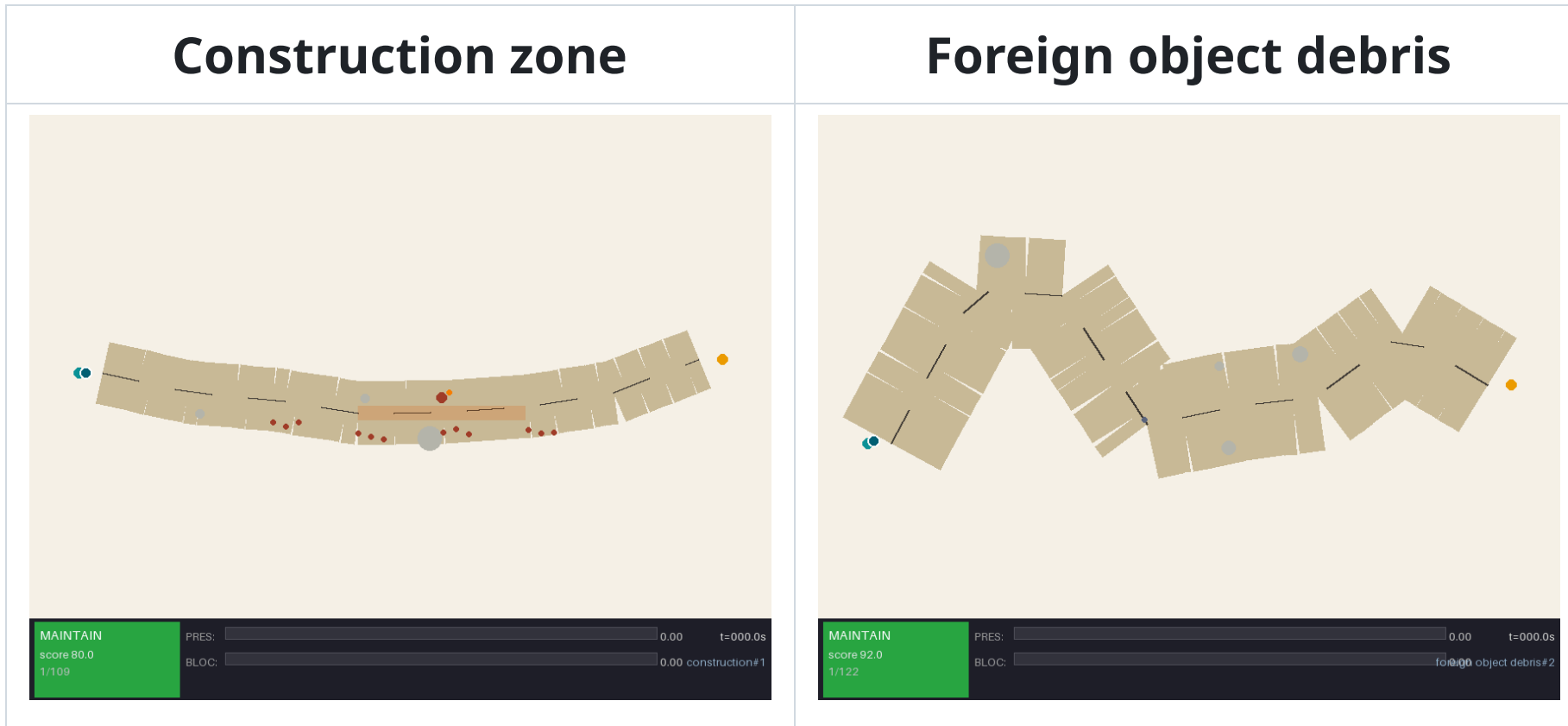
Demo: Minimal-Shot Success And Failure



Same scenario family: wrong-way actor, low visibility, night.

The baseline extrapolates into the hazard. Spotlight selects an evasive token,

More Long-Tail Scenarios



The point is controlled variation of geometry, hazards, clearances, and recovery choices.

Wins Snapshot

Win	Result	Why it matters
closed-loop internal safety	350 rollouts, 0 collisions, 93.1% overall pass	runtime is not a toy one-off demo
baseline gap on hardest internal suite	57.6% vs 2.1% pass, 7.9% vs 20.5% collision	geometry-grounded selection changes behavior materially
real-data grounding signal	7.845 RFS on WOD-E2E validation-CV	frozen scene priors help when the selector head is right

These are three different wins: internal closed-loop competence, strong baseline separation, and non-trivial grounding signal on real WOD-E2E scenes.

Primary Simulation Result

Closed-loop simulation evidence package:

Suite	Runs	Collisions	Pass
WOD-style, all 11 clusters	110	0	100%
Compositional OOD	60	0	100%
Adversarial	60	0	100%
Hidden holdout	60	0	100%
Gauntlet	60	0	60%
Total	350	0	93.1%

COMPASS: **9.137 / 10** over 700 ranked runs.

Source: [README.md](#) | [artifacts/compass_evidence_report.json](#)

Baseline Comparison

Matched gauntlet comparison: same 420 scenarios, same seeds.

Policy	Pass rate	Collision rate
Baseline, no world-state reasoning	2.1%	20.5%
Spotlight Reflex	57.6%	7.9%

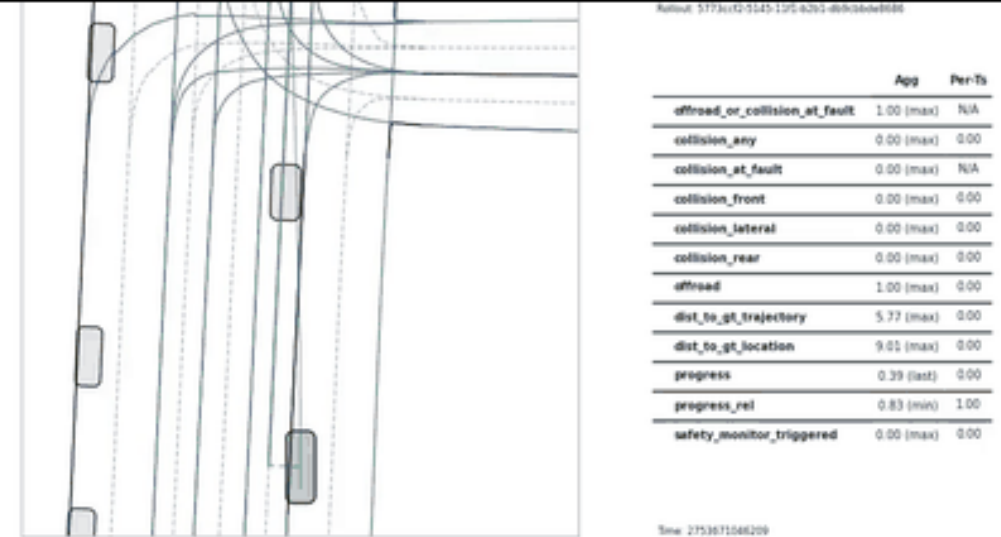
Explicit geometric reasoning gives a large improvement over a non-reasoning baseline in the randomized challenge environment.

AlpaSim External Transfer

This is a WOD-E2E front-camera AlpaSim rollout with adapter map and metrics.

AlpaSim is not used to claim production safety. It is used to test whether the token interface survives a different observation and execution stack.

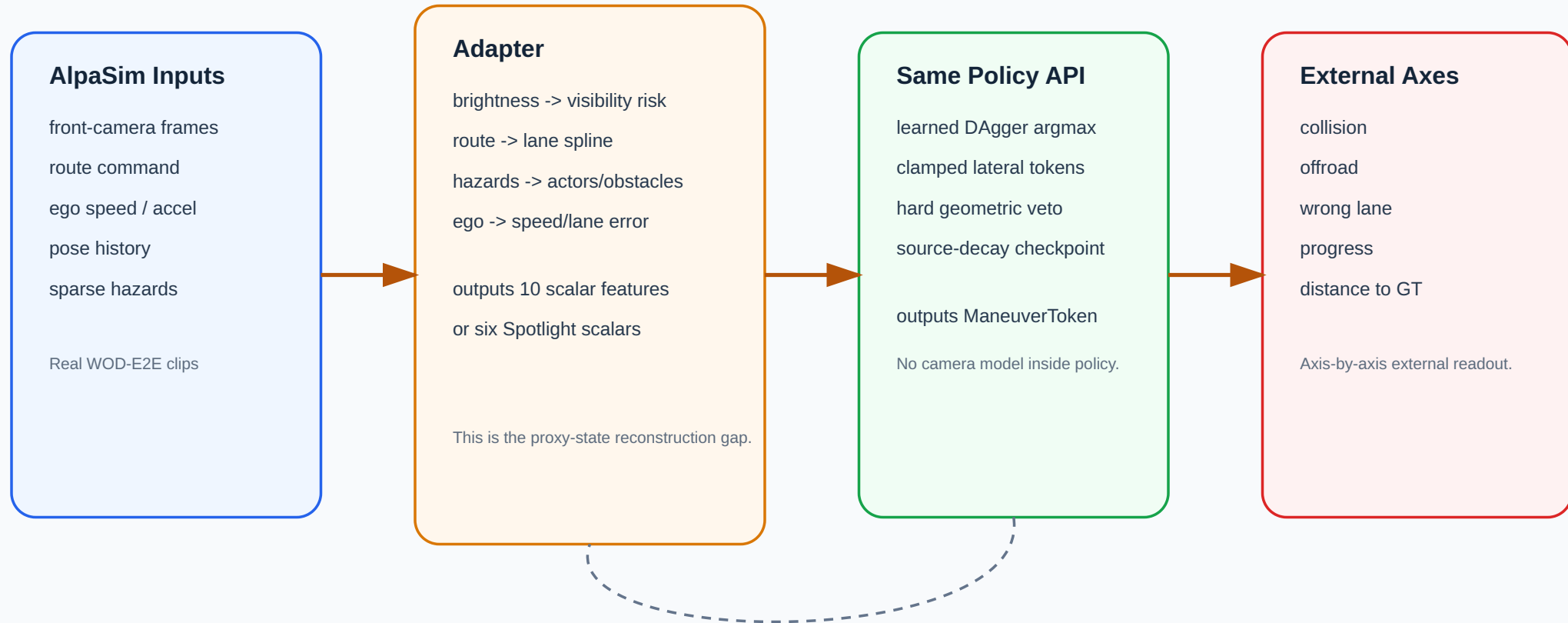
AlpaSim transfer matrix: DAgger token policy on WOD-E2E clip



AlpaSim Adapter

AlpaSim Transfer Stack

The policy is held fixed. Transfer happens through the observation adapter and execution interface.

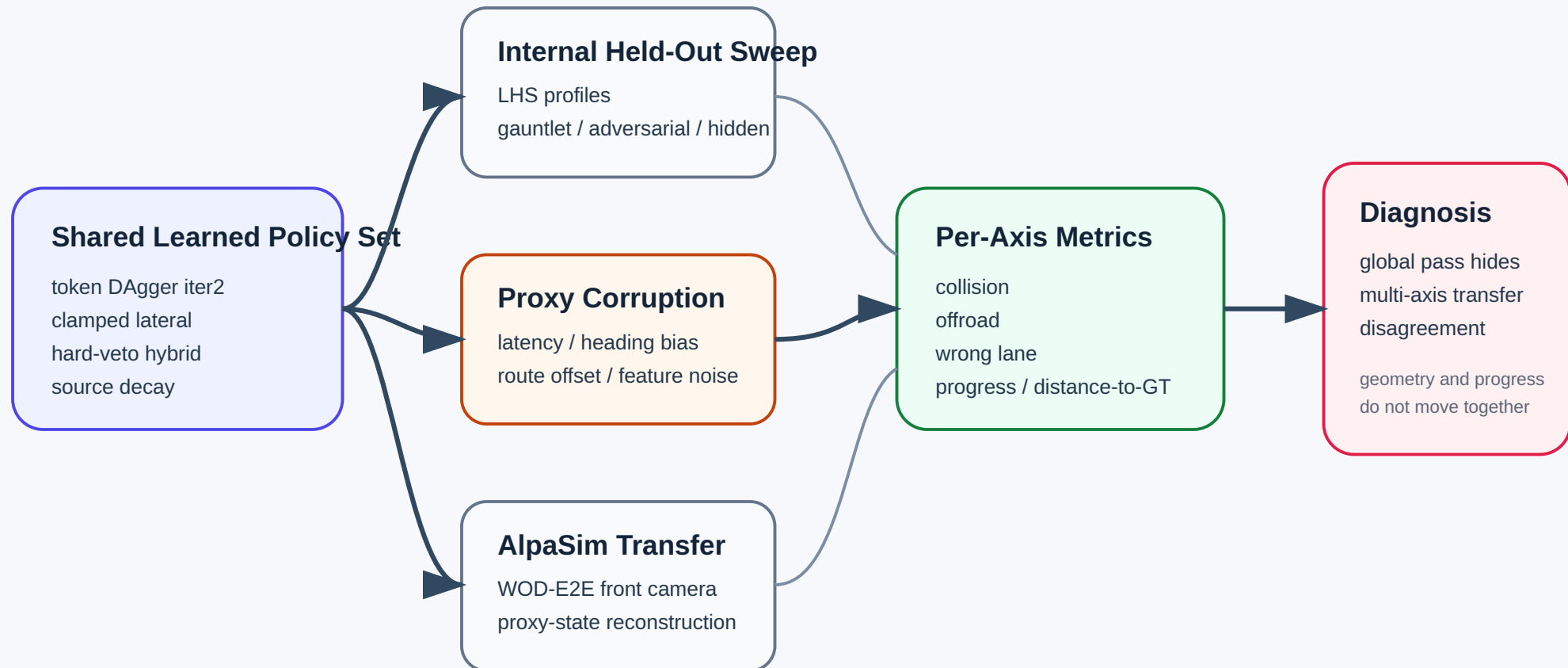


Selection failures appear only after adapter reconstruction and execution.

Transfer Diagnostic Schema

Same policy, three interfaces, different external failures.

Internal success, proxy corruption, and AlpaSim expose different axis tradeoffs.



What Failed Externally

10 matched WOD-E2E clips completed by all four learned variants:

Model	Collision	Offroad	Wrong lane	Progress	Dist. m
Raw DAgger iter2	0.600	0.900	0.700	0.034	61.98
Clamped lateral	0.700	0.500	0.200	0.384	59.85
Hard-veto hybrid	0.800	0.200	0.700	0.837	166.06
Source decay	0.600	0.900	0.400	0.175	62.92

No single learned intervention fixed all axes. This is the main understood failure case: transfer breaks into collision, offroad, lane, and progress axes.

Source: `artifacts/alpasim_matrix10_analysis.md`.

WOD-E2E Auxiliary Track

479 WOD-E2E validation frames, segment-grouped 5-fold CV.

Selector	RFS	Folds	Role
Waymo baseline	7.022	-	reference
Local baseline	7.131	-	reference
Gate-only	7.803	5	selector baseline
RFF direct policy	7.834	5	learned baseline
GPU MLP + Cosmos 64d	7.845	5	best validation-CV
Oracle selector	9.264	5	upper bound

This is auxiliary evidence: the right trajectory often exists, but selection still needs stronger grounded scene understanding.

Reproducibility

Start here:

```
README.md  
presentation.md / presentation-sota.pdf  
docs/corl2027/paper.pdf  
docs/corl2027/AUDIT.md  
./scripts/run_corl2027_audit.sh
```

Primary artifacts:

```
artifacts/minimal_shot_claim_audit.json  
artifacts/alpasim_matrix10_analysis.md  
artifacts/compass_evidence_report.json
```

Branch:

```
https://github.com/amtellezfernandez/minimal-shot-av/tree/CoRL-2027
```

Boundaries

Claim	Status
working minimal-shot AV simulator	yes
autonomous policy demo in randomized long-tail scenes	yes
WOD-E2E validation-CV selector result	yes, not hidden-test
AlpaSim external transfer diagnosis	yes, 10 paired shared clips
production AV safety claim	no
uniform positive learned method	no

This is strongest as an architecture prototype plus a transfer-diagnostic

Research Appendix

The same repo also contains a CoRL 2027 paper draft and full experiment logs.

The paper-focused part of the repo studies one narrower question:

when learned token policies look good internally, what exactly breaks under external transfer, and which failures come from geometry, proxy state, or action ordering?

That analysis lives here because it was built on the same stack, not because this deck is only a paper summary.

Internal Imitation-Learning Result

Held-out Latin-hypercube sweep: 12 profiles, seeds 1-10, Gauntlet / Adversarial / Hidden.

Agent	Overall pass / PV	Gauntlet	Adversarial	Hidden
Continuous-BC	77.69 / 3.52	77.50 / 4.31	75.42 / 2.08	83.33 / 1.67
Token-BC	76.85 / 4.26	77.64 / 4.86	71.67 / 2.92	82.50 / 3.33
Token-RNN-BC	78.43 / 2.96	79.17 / 2.92	74.58 / 2.50	81.67 / 4.17
Token-Dagger-BC, 2-	84.54 / 0.18	96.25 /	80.83 / 0.00	91.67 /

Controlled Proxy-State Test

The simulator reproduces AlpaSim-style tradeoffs when only policy-visible state is corrupted.

Perturbation	Raw p/c/l	Clamp	Hybrid	
clean	0.917/0.083/0.312	0.938/0.062/0.167	0.958/0.042/0.167	0.938,
actor latency	0.604/0.396/0.875	0.604/0.396/0.521	0.854/0.146/0.208	0.833,
route offset	0.938/0.062/0.229	0.875/0.125/0.021	0.896/0.104/0.062	0.938,
feature noise	0.917/0.083/0.917	0.938/0.062/0.146	0.917/0.083/0.125	0.896,

p/c/l means pass / collision / lane-violation rate.

Source: `artifacts/internal_proxy_transfer_medium.md`.

Grounding Context

Recent VLA work argues for physically grounded latent reasoning.

LaST-VLA uses 3D geometric priors and world-model dynamics to improve autonomous-driving VLA planning.

Our complementary lesson:

- grounding is useful
- scalar summaries can hide metric-axis conflicts
- grounded selectors should report collision, lane, offroad, progress, and tracking separately

Reference: [LaST-VLA, arXiv:2603.01928](#)

Final Takeaway

The core contribution is a minimal-shot AV stack that makes the control path, the stress environment, the transfer interface, and the failure modes all explicit.

The strongest claim is concrete:

the stack works internally, transfers unevenly across external axes, and shows exactly where grounding and proxy-state mismatch start to matter.

